



CHALMERS
UNIVERSITY OF TECHNOLOGY



UNIVERSITY OF GOTHENBURG

From Human-to-Human to Human-to-Bot Conversations in Software Engineering

Ranim Khojah, Francisco Gomes de
Oliveira Neto, philipp Leitner



Background

- Chatbots are becoming a part of the development team.
- Chatbots **introduce new aspects to communication** within a team.
- Understanding the dynamics of conversations within a team help:
 - Developers know who to approach based on the situation and what to expect.
 - Chatbot designers understand aspects to consider when designing a chatbot.

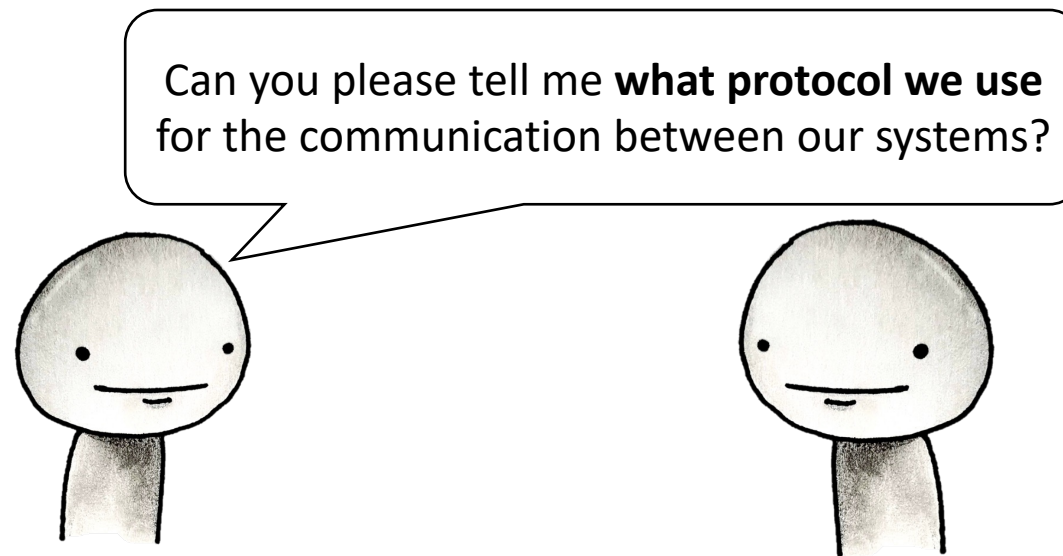


What we did

- Attributes of conversations with humans and NLU-based chatbots
- We extend the comparison to include LLM-based chatbots.
- Purpose, Understanding of scope, Listening, Trust, and Humour.
- Conversations with **Human Developers**, **NLU-Chatbots** and **LLM-Chatbots**

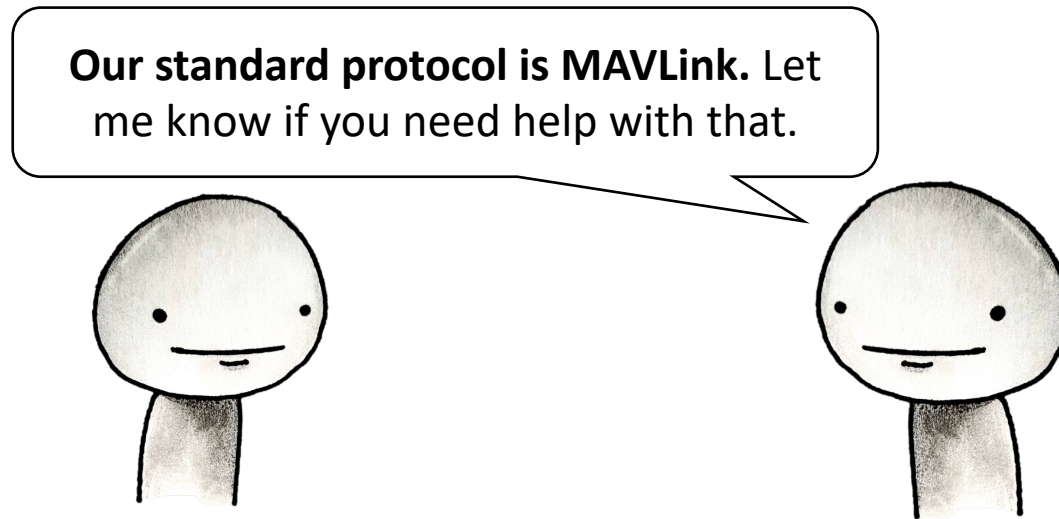
Conversations with Human Developers

- Social purposes or to receive information for guidance or training.
- Trust based on previous interactions and shared moments.
- Conversations are **built on mutual understanding**.



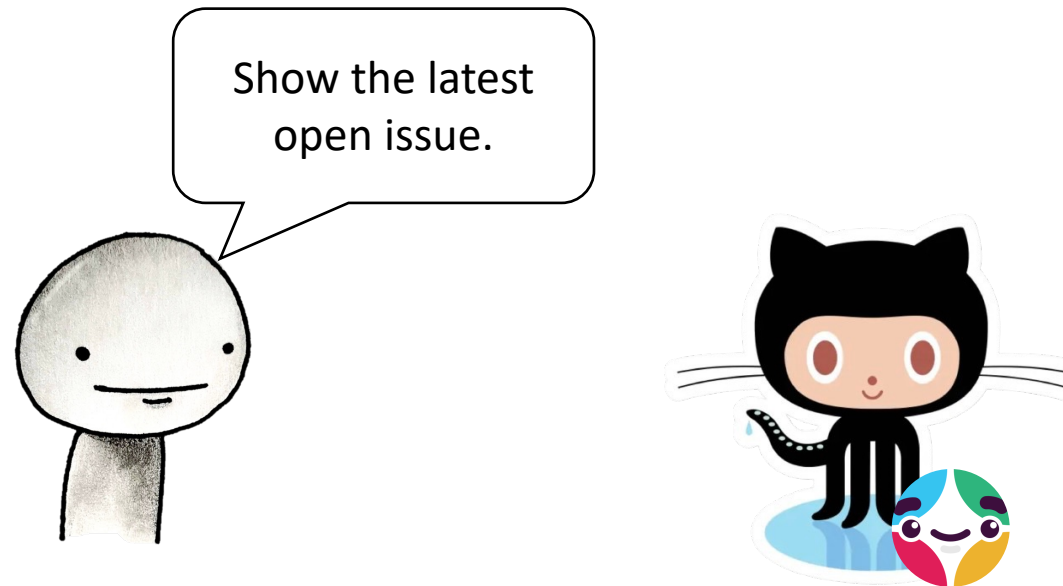
Conversations with Human Developers

- Active listening through body language, while **accurate listening requires knowledge** and expertise.
- Humor is mostly acceptable



Conversations with NLU-based Chatbot

- For basic information retrieval and simple automation.
- **Customized during the design** to align with the user's preferences.



Conversations with NLU-based Chatbot

- Intent classification to accurately listen to the developer.
- Trust is impacted by the **performance** and an efficient automation.

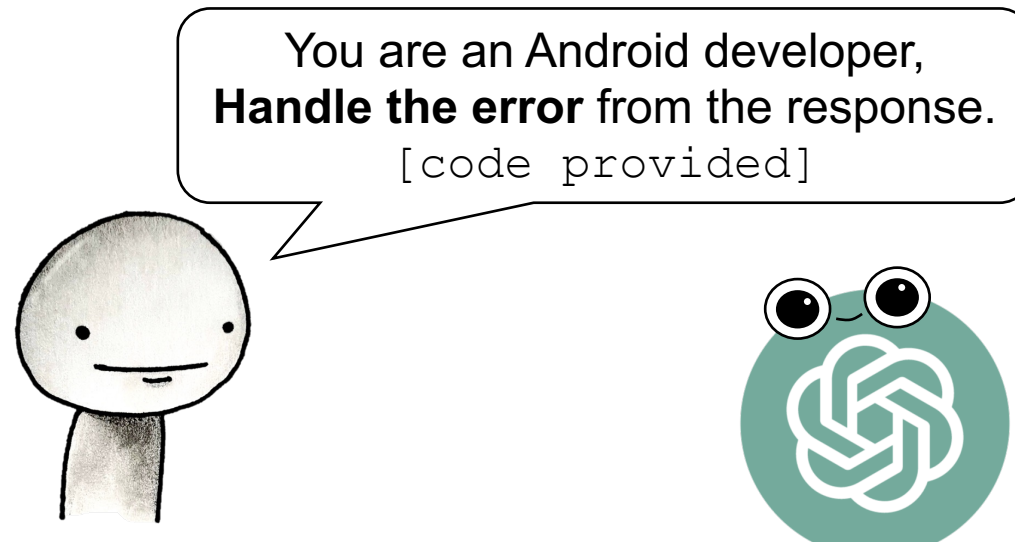


Issue #4: Vulnerability in the authorization

Authentication error response not handled properly.

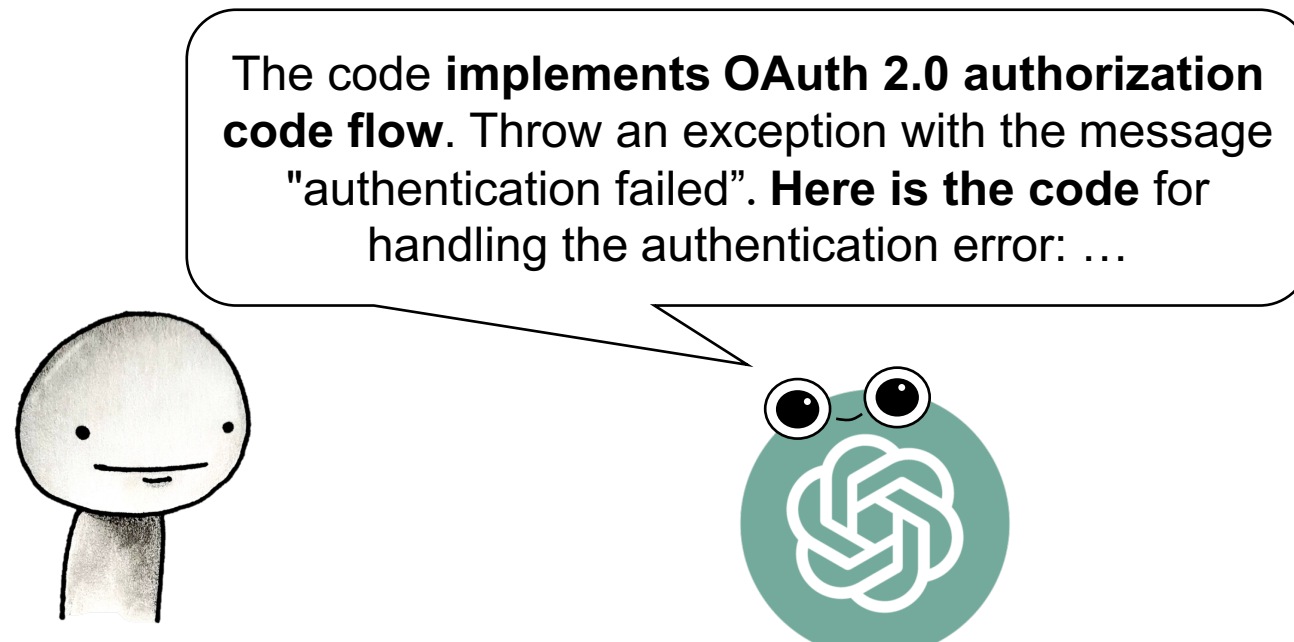
Conversations with LLM-based Chatbot

- Artifact manipulation, general guidance, or training
- Additional option for **customization during the interaction.**
- Developers trust LLM-chatbots that meet their **expectations.**



Conversations with LLM-based Chatbot

- LLM-chatbots listen by receiving the prompt, summarizing the query then responding with a contextually-relevant response.



Lessons learned

- Trustworthiness attribute mainly **determines the flow** of a conversation.
- LLM-based chatbots enable human-like conversations, but with bot-alike efficiency.
- Conversation styles are not mutually exclusive, but rather complementary.

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Connection to GF

- Support for a wider grammar that is present in different conversational styles.
 - E.g., “As a software developer, I want you to develop a game.”
- GF can improve trust in chatbots
 - E.g., improved intent classification.

